

Classical Encryption Techniques

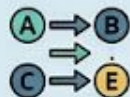
(Substitution and Transposition techniques)

TYPES OF CLASSICAL ENCRYPTION TECHNIQUES

Classical encryption methods are divided into two primary categories.

CLASSICAL ENCRYPTION

SUBSTITUTION CIPHER



REPLACES CHARACTERS:
e.g., A becomes D, B becomes E

1	2	3	4
2			
3		L	
4			

POSITION REMAINS SAME:
Character sequence order is preserved.



FREQUENCY CHANGES:
Letter frequency distribution is modified.

TRANSPOSITION CIPHER



REARRANGES CHARACTERS:
e.g., characters are scrambled.

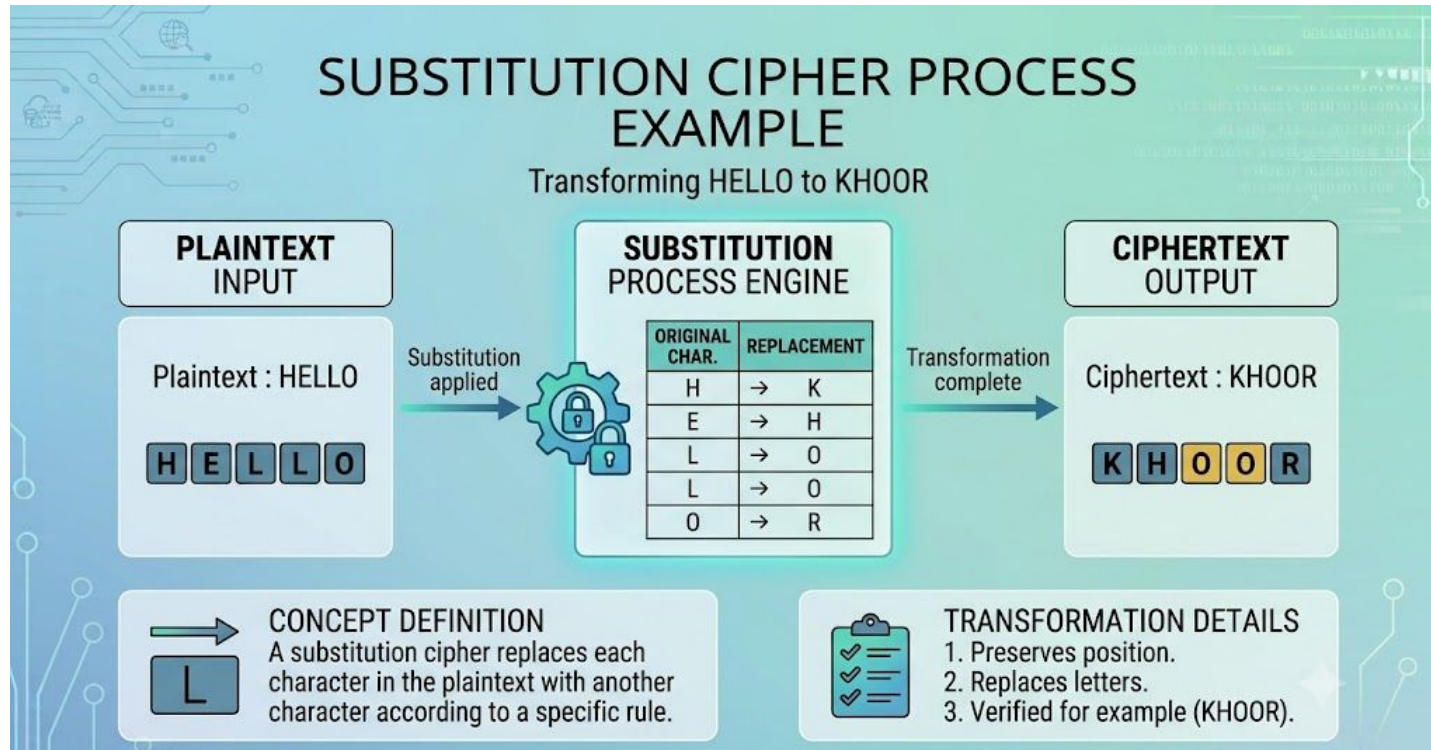
1	2	3	4
2			
3		L	
4			

CHARACTER POSITIONS SHIFT:
Characters remain the same set, but positions jumble.



FREQUENCY REMAINS SAME:
Preserves the character set and frequency.

Substitution Technique



Types of Substitution Ciphers

1. Caesar Cipher
2. Monoalphabetic Cipher
3. Playfair Cipher
4. Hill Cipher
5. Vigenère Cipher

Types of Ciphers

1. Classical Ciphers

Used in ancient and historical cryptography.

- Caesar Cipher
- Shift Cipher
- Monoalphabetic Cipher
- Playfair Cipher
- Vigenère Cipher
- Rail Fence Cipher
- Columnar Transposition Cipher

2. Modern Ciphers

Used in today's secure communication.

- AES (Advanced Encryption Standard)
- DES (Data Encryption Standard)
- Triple DES (3DES)
- RSA
- ECC (Elliptic Curve Cryptography)

Why Do We Use Ciphers?

- Protect confidential information
- Secure online banking and payments
- Secure passwords
- Protect emails and messages
- Enable secure internet communication (HTTPS, VPN)

Real-Life Examples

Application	Cipher Used
WhatsApp Messages	AES + Signal Protocol
HTTPS Websites	AES, RSA, ECC
Wi-Fi Security (WPA2/WPA3)	AES
ATM Transactions	AES, Triple DES
Digital Signatures	RSA, ECC

Substitution Technique

★ Letters are replaced by other letters or symbols.

Example:

a	b	c	d	e	f	g	h	i	j	k	l	m
n	o	p	q	r	s	t	u	v	w	x	y	z

a → M

b → X

x → Z

g → A

Plaintext : bag

Ciphertext : XMA

Transposition Technique

- ★ Applying some sort of permutation on the plaintext letters.
- ★ Plaintext: NESO
- ★ Ciphertext: ESON, SONE, ONES, ENOS

Classical Encryption Technique

Substitution	Transposition
❖ Caesar Cipher ❖ Monoalphabetic Cipher ❖ Playfair Cipher ❖ Hill Cipher ❖ Polyalphabetic Cipher ❖ One-Time Pad	❖ Rail Fence ❖ Row Column Transposition

Caesar Cipher

- ★ Letters are replaced by other letters or symbols.
- ★ The earlier known and simplest method used by Julius Caesar.
- ★ Replacing each letter of the alphabet with the letter standing three places further down the alphabet.

Example:

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher

Algorithm:

For each plaintext letter 'p', substitute the ciphertext letter 'C':

$$C = E(p, k) \bmod 26 = (p + k) \bmod 26$$

$$p = D(C, k) \bmod 26 = (C - k) \bmod 26$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y

$$\begin{array}{r} 0 \\ 26 \overline{) 16} \\ 0 \\ \hline 16 \end{array}$$

$$C = (p + k) \bmod 26$$

$$C = (13 + 3) \bmod 26$$

$$C = 16 \bmod 26$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y

$$C = (p + k) \bmod 26$$

$$C = (13 + 3) \bmod 26$$

$$C = 16 \bmod 26$$

$$C = 16$$

$$C = Q$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q										

$$C = (p + k) \bmod 26$$

$$C = (4 + 3) \bmod 26$$

$$C = 7 \bmod 26$$

$$C = 7$$

$$C = H$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H									

$$C = (p + k) \bmod 26$$

$$C = (18 + 3) \bmod 26$$

$$C = 21 \bmod 26$$

$$C = 21$$

$$C = V$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M

13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	B

$$C = (p + k) \bmod 26$$

$$C = (24 + 3) \bmod 26$$

$$C = 27 \bmod 26$$

$$C = 1$$

$$C = B$$

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Example

Question: Encrypt “neso academy” using Caesar cipher.

Solution:

n	e	s	o	a	c	a	d	e	m	y
Q	H	V	R	D	F	D	G	H	P	B

Plaintext: nesoacademy

Ciphertext: QHVRDFDGHPB

Homework!

Decrypt "QHWZRUN VHFUXULWB" using Caesar cipher.

Shift Cipher

Key = 2, 3, 4, 5, ...

Shift Cipher with Key = 3 is called Caesar Cipher.

Example:

Plaintext : Neso

Key : 4

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Shift Cipher

Key = 2, 3, 4, 5, ...

Shift Cipher with Key = 3 is called Caesar Cipher.

Example:

Plaintext : Neso

Key : 4

Ciphertext: RIWS

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Caesar Cipher – Pros and Cons

Pros

1. Simple
2. Easy to implement.

Cons

1. The encryption and decryption algorithms are known.
2. There are only 25 keys to try. (Vulnerable to Brute-force attack)
3. The language of the plaintext is known and easily recognizable.

Brute force attack

Ciphertext: SQDYMZK

Shifts	Back	Result
--------	------	--------

0	[26]	SQDYMZK
1	[25]	TREZNAL
2	[24]	USFAOBM
3	[23]	VTGBPCN
4	[22]	WUHCQDO
5	[21]	XVIDREP
6	[20]	YWJESFQ
7	[19]	ZXKFTGR
8	[18]	AYLGUHS
9	[17]	BZMHVIT
10	[16]	CANIWJU
11	[15]	DBOJXKV
12	[14]	ECPKYLW
13	[13]	FDQLZMX

Shifts	Back	Result
--------	------	--------

13	[13]	FDQLZMX
14	[12]	GERMANY
15	[11]	HFSNBOZ
16	[10]	IGTOCPA
17	[9]	JHUPDQB
18	[8]	KIVQERC
19	[7]	LJWRFSO
20	[6]	MKXSGTE
21	[5]	NLYTHUF
22	[4]	OMZUIVG
23	[3]	PNAVJWH
24	[2]	QOBWKXI
25	[1]	RPCXLYJ

Brute force attack

Ciphertext: SQDYMZK

Shifts	Back	Result
0	[26]	SQDYMZK
1	[25]	TREZNAL
2	[24]	USFAOBM
3	[23]	VTGBPCN
4	[22]	WUHCQDO
5	[21]	XVIDREP
6	[20]	YWJESFQ
7	[19]	ZXKFTGR
8	[18]	AYLGUHS
9	[17]	BZMHVIT
10	[16]	CANIWJU
11	[15]	DBOJXKV
12	[14]	ECPKYLW
13	[13]	FDQLZMX

Shifts	Back	Result
13	[13]	FDQLZMX
14	[12]	GERMANY
15	[11]	HFSNBOZ
16	[10]	IGTOCPA
17	[9]	JHUPDQB
18	[8]	KIVQERC
19	[7]	LJWRFSO
20	[6]	MKXSGTE
21	[5]	NLYTHUF
22	[4]	OMZUIVG
23	[3]	PNAVJWH
24	[2]	QOBWKXI
25	[1]	RPCXLYJ

Homework!

1. Encrypt the plaintext "cryptography" using shift cipher (Key=4).
2. Encrypt "Neso" using shift cipher (Key=0).



Monoalphabetic Cipher

Introduction

- A monoalphabetic cipher is a type of **simple substitution cipher** in which **each letter of the plaintext is replaced with a fixed letter**.
- In a monoalphabetic cipher, we can replace a single character with **any other 25 characters**, and **the order of replacement is not fixed**.
- For example, **A** can be replaced with **any character from B to Z**, and **B** can be replaced with **A or C to Z**, and so on.

- Unlike the Caesar Cipher, the substitution is **not based on a fixed shift**. Instead, a random permutation of the alphabet is used as the encryption key.
- Since each letter can be mapped to any one of the remaining 25 letters, there are **26! (26 factorial)** possible substitution keys, making it more secure than the Caesar Cipher.
- However, it is still vulnerable to **frequency analysis** because the same plaintext letter is always replaced by the same ciphertext letter.

Monoalphabetic Cipher – Problem

- Let's take an example to understand Monoalphabetic Cipher
- Encrypt plaintext **ATTACK**
- Using the substitution key (mapping) below:

Plaintext Alphabet	A	B	C	D	E	F	G	H	I	J	K	M
Ciphertext Alphabet	Q	W	E	R	T	Y	U	I	O	P	A	S
Plaintext Alphabet	N	O	P	Q	R	S	T	U	V	W	X	Y
Ciphertext Alphabet	F	G	H	J	K	L	Z	X	C	V	B	M

Monoalphabetic Cipher – Step-by-Step Encryption

- Let's now **encrypt** the word "ATTACK" letter by letter:

Plaintext Alphabet	A	B	C	D	E	F	G	H	I	J	K	L	M
Ciphertext Alphabet	Q	W	E	R	T	Y	U	I	O	P	A	S	D
Plaintext Alphabet	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Ciphertext Alphabet	F	G	H	J	K	L	Z	X	C	V	B	N	M

Plaintext Letter	A	T	T	A	C	K
Lookup in Substitution Key	A → Q	T → Z	T → Z	A → Q	C → E	K → A
Ciphertext Letter	Q	Z	Z	Q	E	A

Monoalphabetic Cipher

- ★ The “cipher” line can be any permutation of the 26 alphabetic characters.

Permutation

- ★ A permutation of a finite set of elements 'S' is an ordered sequence of all the elements of 'S', with each element appearing exactly once.
- ★ For example $S = \{ a , b , c \}$
- ★ There are six permutations of S.

abc

acb

bac

bca

cab

cba

Monoalphabetic Cipher

- ★ The “cipher” line can be any permutation of the 26 alphabetic characters.
- ★ This would seem to eliminate brute-force techniques for cryptanalysis.
- ★ A single cipher alphabet (mapping from plain alphabet to cipher alphabet) is used per message.
- ★ English language - Nature of plain text is known.

Monoalphabetic Cipher

O	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Monoalphabetic Cipher

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z



Example: Plain Text - Neso

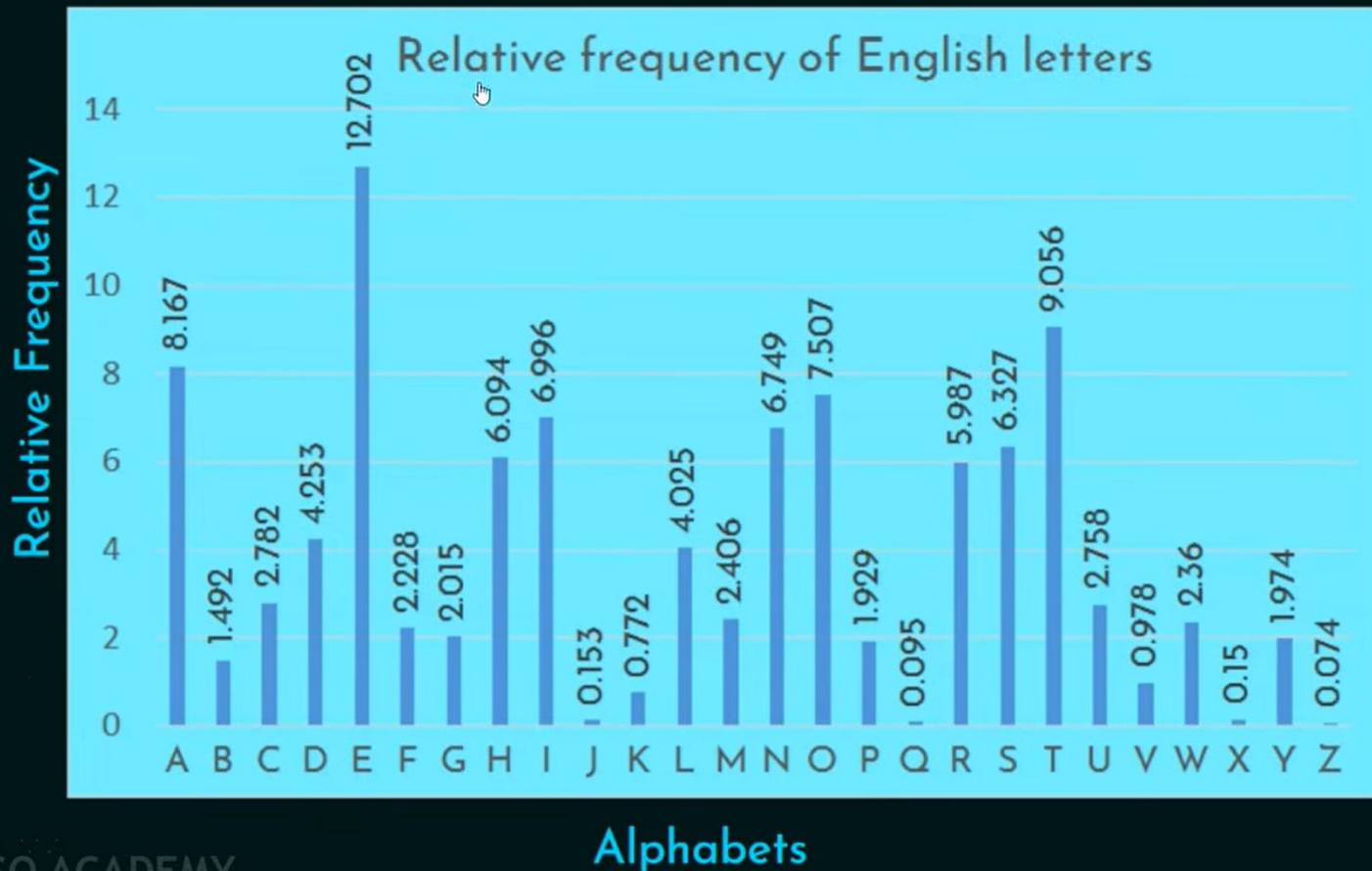
Monoalphabetic Cipher

0	1	2	3	4	5	6	7	8	9	10	11	12
A	B	C	D	E	F	G	H	I	J	K	L	M
13	14	15	16	17	18	19	20	21	22	23	24	25
N	O	P	Q	R	S	T	U	V	W	X	Y	Z

Example: Plain Text - Neso

Cipher Text - DULA

Relative frequency of English letters



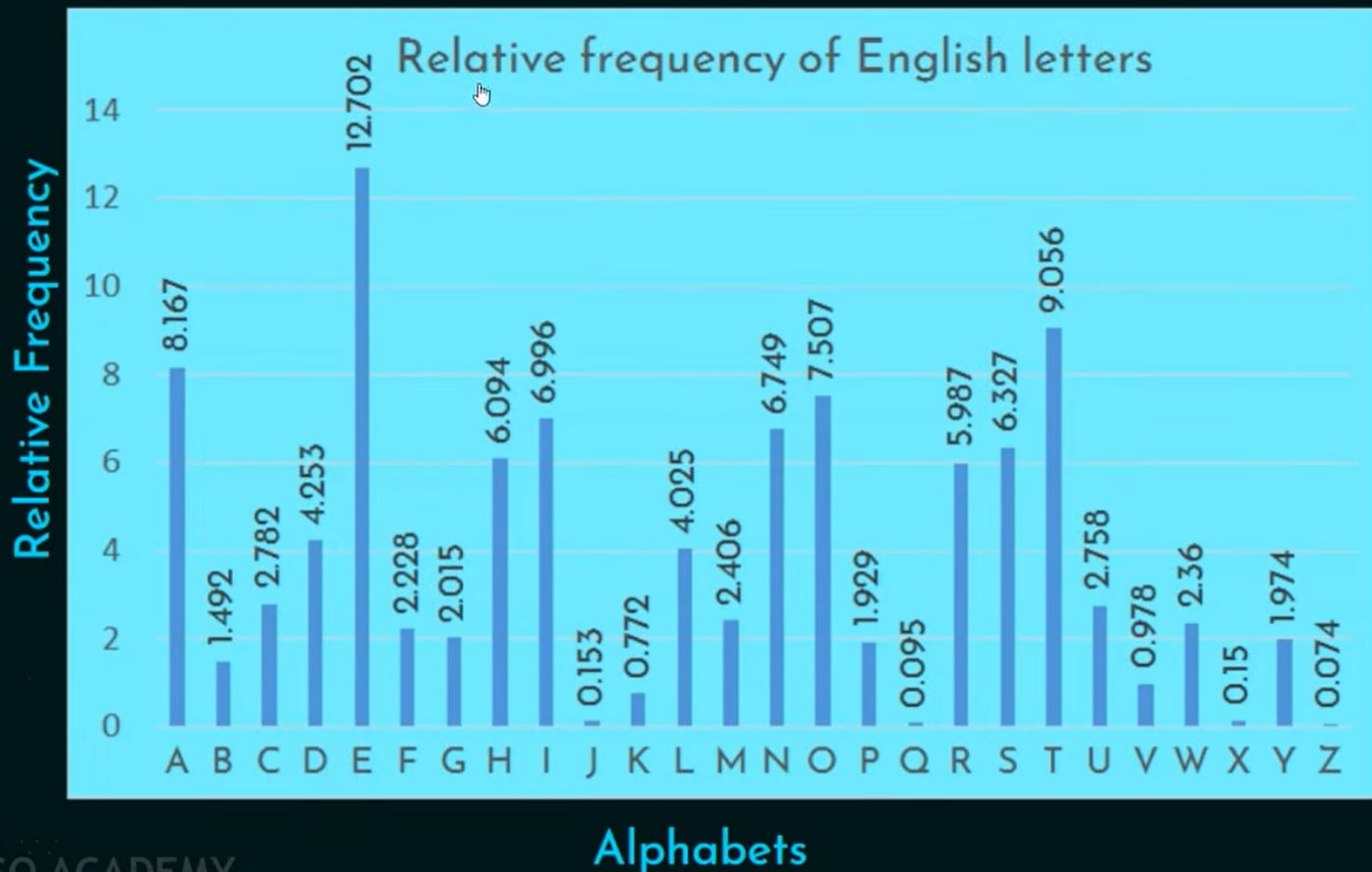
Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

Monoalphabetic Cipher – Example

P 13.33	E 5.00	B 1.67	N 0.00
Z 11.67	V 4.17	G 1.67	R 0.00
S 8.33	X 4.17	Y 1.67	
U 8.33	F 3.33	I 0.83	
O 7.50	W 3.33	J 0.83	
M 6.67	Q 2.50	C 0.00	
H 5.83	T 2.50	K 0.00	
D 5.00	A 1.67	L 0.00	

Relative frequency of English letters



Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
----	---	---	---	---	---	---	---	---	---	---	---

Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
PT	E		E				E				
PT	E		E			T	E				
PT	E		E			T	E			A	
PT	E		E			T	E		L	A	N
PT	E		E			T	E	P	L	A	N

Monoalphabetic Cipher – Example

GZGEWVGRNCP

CT	G	Z	G	E	W	V	G	R	N	C	P
PT	E		E				E				
PT	E		E			T	E				
PT	E		E			T	E			A	
PT	E		E			T	E		L	A	N
PT	E		E			T	E	P	L	A	N
PT	E	X	E	C	U	T	E	P	L	A	N

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
t a e e te a that e e a a
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
e t ta t ha e ee a e th t a
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ
e e e tat e the t

Monoalphabetic Cipher – Example

UZQSOVUOHXMOPVGPOZPEVSGZWSZOPFPESXUDBMETSXAIZ
VUEPHZHMDZSHZOWSFPAPPDTSVPQUZWYMXUZUHSX
EPYEPOPDZSZUFPOMBZWPFUPZHMDJUDTMOHMQ

it was disclosed yesterday that several informal but
direct contacts have been made with political
representatives of the viet cong in moscow

Monoalphabetic Cipher – Pros and Cons

Pros

1. Better security than Caesar cipher.

Cons

1. Monoalphabetic ciphers are easy to break because they reflect the frequency data of the original alphabet.
2. Prone to guessing attack using the English letter frequency of occurrence of letters.
3. A countermeasure is to provide multiple substitutes, known as homophones, for a single letter.

Homework!

Encrypt the plaintext "Attack postponed to tomorrow and do not use our secret paper until further info" using Monoalphabetic cipher technique.

Secret Key: The quick brown fox jumps over the lazy dog.

Note: Ignore the second and latter occurrence of alphabets in the key.

Playfair Cipher

Playfair Cipher

- ★ Aka Playfair square or Wheatstone-Playfair cipher.
- ★ Manual symmetric encryption technique.
- ★ The first literal digram substitution cipher.
- ★ Invented in 1854 by Charles Wheatstone.
- ★ Bore the name of Lord Playfair for promoting its use.

Playfair Cipher

- ★ Multiple letter encryption cipher.
- ★ Digrams.
- ★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

M	O	N	A	R
C	H	Y		

Playfair Cipher

- ★ Multiple letter encryption cipher.
- ★ Digrams.
- ★ 5 x 5 matrix constructed using a keyword (Ex: Monarchy)

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇔ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

Plaintext: attack

Digrams: at ta ck

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

Plaintext: attack

Digrams: at ta ck

Plaintext: neso academy

Digrams: ne so ac ad em yx

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

Plaintext: attack

Digrams: at ta ck

Plaintext: neso academy

Digrams: ne so ac ad em yx

Plaintext: balloon

Digrams: ba ll oo n

Digrams: ba lx lo on

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇌ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

at	ta	ck

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇌ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

at	ta	ck
RS		

M	O	N	A → R	
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S ← T	
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

at	ta	ck
RS	SR	

M	O	N	A → R	
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S ← T	
U	V	W	X	Z

Understanding the rules

Example 1: attack

Digrams: at ta ck

at	ta	ck
RS	SR	DE

Plaintext: attack

Digrams: RSSRDE

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 2: mosque

mo	sq	ue

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rules for encryption using Playfair Cipher

1. Digrams.
2. Repeating Letters - Filler letter.
3. Same Column | ↓ | Wrap around.
4. Same row | → | Wrap around.
5. Rectangle | ⇌ | Swap

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 2: mosque

mo	sq	ue
ON		

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 2: mosque

mo	sq	ue
ON	TS	

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Understanding the rules

Example 2: mosque

mo	sq	ue
ON	TS	ML

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z



Understanding the rules

Example 2: mosque

mo	sq	ue
ON	TS	ML

Plaintext: mosque

Digrams: ONTSML

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Rapid Fire – Activity

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Digrams

oa

ab

gk

dc

rz

zr

mz

rm



Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Plaintext : Hide the gold under the carpet

Key : Neso Academy

Ciphertext : IKGDQKDPNRCVE COPQKNDOTVDRZ

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ



N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

→
H → I/J → I
I → K

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	N D	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

↓
D → G E →

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ



N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

$T \rightarrow Q$
 $H \rightarrow K$

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E ✓	S	O	A
C	D ✓	M	Y	B
F	G ✓	H	I/J	K
L	P ✓	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ



N✓	←E	S	→O✓	A
C	D	M	Y	B
F	G	H	I/J	K
L✓	P	Q	→R	T
U	V	W	X	Z

O → N
L → R

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C ← D ✓		M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U → V		W	X	Z

D → C
U → V

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ



N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ



N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	ND	OT	VD	RZ

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	N D	OT	VD	RZ
N	E	S	O	A								
C	D	M	Y	B								
F	G	H	I/J	K								
L	P	Q	R	T								
U	V	W	X	Z								

Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Plaintext : Hide the gold under the carpet

Key : Neso Academy

Ciphertext : IKGDQKDPNRCVE COPQKNDOTVDRZ



Solution

Hi	de	th	eg	ol	du	nd	er	th	ec	ar	pe	tx
IK	GD	QK	DP	NR	CV	EC	OP	QK	N	OT	VD	RZ
									D			✓

N	E	S	O	A
C	D	M	Y	B
F	G	H	I/J	K
L	P	Q	R	T
U	V	W	X	Z

Question

Encrypt the message "Hide the gold under the carpet" using playfair technique with the keyword "Neso Academy".

Plaintext : Hide the gold under the carpet

Key : Neso Academy

Ciphertext : IKGDQKDPNRCVECOPOKNDOTVDRZ



Homework

Decrypt the ciphertext "ODZFQSEZSONTSW" using playfair technique with the keyword "Neso App".